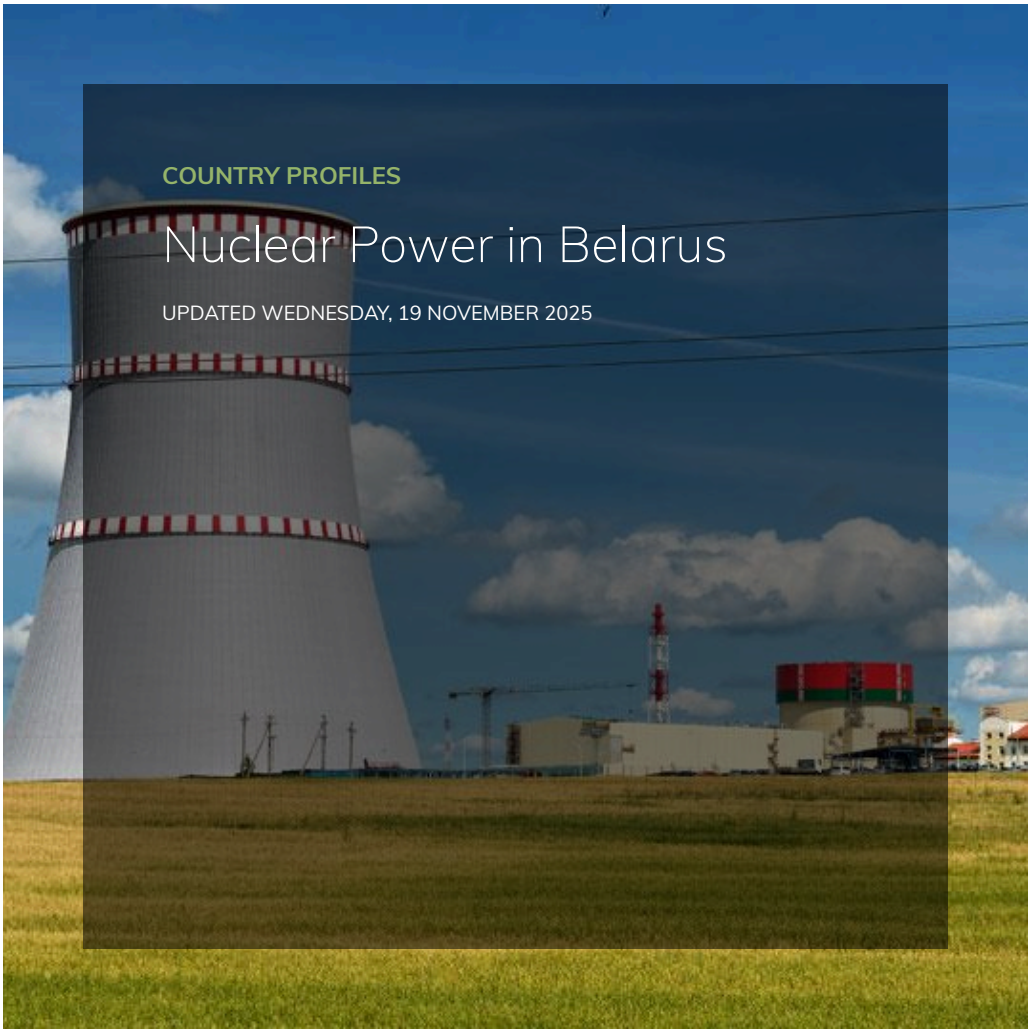


COUNTRY PROFILES

Nuclear Power in Belarus

UPDATED WEDNESDAY, 19 NOVEMBER 2025

- Belarus has one nuclear power plant at Ostrovets. In November 2020 the first unit was connected to the grid, with the second unit connected in May 2023. The project is financed by Russia.
- The Ostrovets project is financed by Russia and the two VVER-1200 units were built by Atomstroyexport.
- In November 2025 Belarus decided to build a third unit at Ostrovets and begin surveys for a potential second site in Mogilev.

Privacy - Terms

2

OPERABLE REACTORS

2,220 MWe

0

REACTORS UNDER CONSTRUCTION

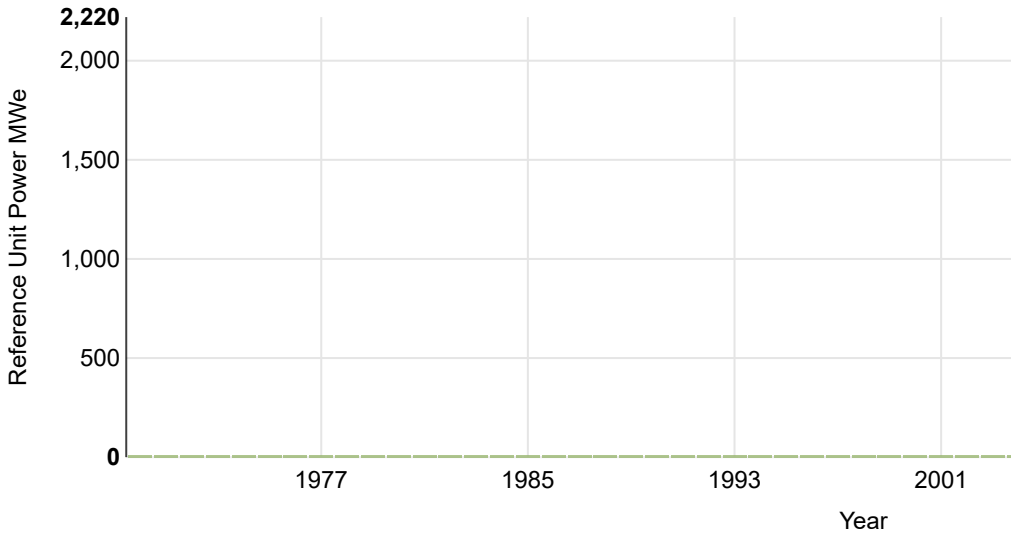
0 MWe

0

REACTORS SHUTDOWN

0 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 41.7 TWh

Generation mix: natural gas 27.0 TWh (65%); nuclear 11.8 TWh (28%); oil 1.7 TWh (4%); biofuels 0.4 TWh.

Import/export balance: 0.3 TWh net export (4.2 TWh imports; 4.5 TWh exports)

Total consumption: 32.8 TWh

Per capita consumption: c. 3600 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Government plans to reform the electricity sector by creating a wholesale market in three stages have stalled, and electricity remains heavily subsidized for households.

Under its 2011-2020 energy strategy, Belarus set out to reduce its reliance on Russia as its single supplier of natural gas. The nuclear power plant under construction at Ostrovets is pivotal to the plan, which also outlined the proposed building of a 1000 MWe coal-fired plant, four hydropower stations with a total capacity of 120 MW, and wind projects totaling 300 MW.

The country imports all of its gas from Russia – much of it for electricity. Gas imports cost about $2.3 billion in 2020[a]. The under construction nuclear plant is expected to reduce gas imports by 5 billion m^3 per year. In November 2011 it was agreed that Russia's Gazprom would pay $2.5 billion for the 50% of Belarus' gas transmission network, Beltransgaz, that it did not already own. This was linked both to lower gas prices and to Russian finance for the nuclear plant. Earlier, there had been studies on both a domestic nuclear power plant using Russian technology, and Belarus participation in a new nuclear unit at Smolensk or Kursk in Russia.

Nuclear power plants

Operable reactors in Belarus

Reactor name	Model	Reactor type	Net capacity (MWe)	Construction start	Grid connection
Ostrovets 1	VVER V-491	PWR	1110	2013-11-08	2020-11-03
Ostrovets 2	VVER V-491	PWR	1110	2014-04-27	2023-05-13

Source: World Nuclear Association

In mid-2006 the government approved a plan for the construction of an initial 2000 MWe PWR nuclear power plant in the Mogilev region of eastern Belarus. This was expected to provide electricity at half the cost of that from Russian gas and to provide some 30% of the country's electricity by 2020 at a cost of about €4 billion (January 2008 estimate) on a turnkey basis.

After expressions of interest from international reactor vendors were invited, the energy ministry announced in August 2008 that proposals had been received from Atomstroyexport, Westinghouse-Toshiba and Framatome (now Orano). Anything from the USA would need several years for an intergovernmental agreement, and Framatome's EPR was noted as being too big for the first plant. In addition, the energy ministry received a proposal from China Guangdong Nuclear Power Corporation (CGNPC, now China General Nuclear Power Corporation, CGN). Russia's Atomstroyexport emerged as the most likely supplier for the 2 x 1000 MWe plant since the others either did not provide all the information required or could not build the plant soon enough. Operation of the first unit was initially envisaged for 2016 and the second in 2018, and two further units were proposed for operation by 2025. In June 2007 Russia's Eximbank offered a $2 billion credit line to enable purchase of equipment from Russia's Power Machines company as a major part of the overall cost.

In November 2007 a presidential decree defined the organizations responsible for preparing for the construction of the country's first nuclear power plant and budgeted money for engineering and site selection. The candidate sites were Krasnopolyansk and Kukshinovsk (both in the Mogilev region) and Ostrovets in the Grodno region. Ostrovets/Astravets, 23 km from the Lithuanian border and 55 km from Vilnius, was chosen in December 2008, despite protests from Lithuania.

The Directorate for Construction of Nuclear Power Plants was established under the Ministry of Energy. The state-run Belnipienergoprom enterprize was designated as the general designer of the plant with responsibility for negotiating and signing contracts, carrying out feasibility studies and preparing tender documents.

Ostrovets nuclear plant

In June 2009 the government announced that Atomstroyexport would be the general contractor, with Russian and Belarus subcontractors, notably St Petersburg Atomenergoproekt. An intergovernmental agreement concerning the plant was signed in March 2011. A preliminary turnkey construction contract with Atomstroyexport for a 2400 MWe plant (2 x 1200 MWe AES-2006 units using V-491 reactors) was signed in October 2011 by Belarus state-owned Nuclear Power Plant Construction Directorate, and a general construction contract was signed in July 2012. St Petersburg AEP (now part of ASE Group) has been involved with the project since 2004, including site selection and technology choice. In January 2014 the Nuclear Power Plant Construction Directorate became the Belarus Nuclear Power Plant state unitary enterprise.

Also, in June 2009 the government announced that $9 billion Russian financing had been lined up, and Belarus also wanted the Russian loan to include provision for infrastructure. Belarus official cost estimate including infrastructure was $9.4 billion, with one third of this to be spent 2011-15. In November 2011 it was agreed that Russia would lend up to $10 billion for 25 years to finance 90% of the contract between Atomstroyexport and the Belarus Directorate for Nuclear Power Plant Construction (now the Belarus NPP state unitary enterprise). In February 2012 Russian state-owned Vnesheconombank (VEB) and Belarusian commercial bank Belvnesheconombank (BelVEB) signed an agreement needed to implement the Russian export credit facility. In May 2012 the parties said that the first instalment under the design contract would be $204 million, and that this would be followed by $285

million for pre-construction site works. This was confirmed with an agreement signed in May 2014.

Earlier in December 2011 the Nuclear Power Engineering Department of the Energy Ministry submitted an application for a construction licence for the Ostrovets plant to the state nuclear regulator, the Department for Nuclear and Radiation Safety (Gosatomnadzor) of the Ministry of Emergencies.

In April 2013 Atomenergomash (AEM) won the tender to supply the two reactors, which were being manufactured by AEM-Technologies at the Atommash plant at Volgodonsk. In October 2013 Gosatomnadzor (GAN) issued the Nuclear Power Plant Construction Directorate (Belarus AEC) with a licence for the construction of the first reactor at the Ostrovets site. First concrete was in November 2013, with VNIPIET (formerly SPbAEP) as main contractor, though the full construction licence was not issued until April 2014. A construction licence for the basemat of unit 2 was issued by the ministry in February 2014. Construction of unit 2 started in May 2014, several months ahead of schedule. The full construction licence was issued at the end of December 2014.

The reactor vessel for Ostrovets 1 during manufacture (AEM-Technology)

In September 2015 the prime minister ordered relevant ministries and the Minsk executive to allocate the required number of qualified workers to “eliminate delays to construction and installation work” so that the project remained on schedule. In July 2016 the unit 1 reactor pressure vessel (RPV) slipped out of its sling when being moved and hit the ground, apparently without being

damaged. OKB Gidropress and Atomstroyexport in August inspected the RPV and sent the results of that to Belarusian Nuclear Power Plant Company – and to Gosatomnadzor. However, the Belarus government said that it wanted it replaced, and Rosatom responded that it would do so, later making it clear that this was a public acceptance matter rather than a technological issue implying any safety issue. The RPV for unit 2 is now being used in unit 1, incurring about a six-month delay in construction but no extra cost for the customer, and the original RPV will be used elsewhere. The new one was shipped from Atommash in Volgodonsk late in October 2016, and in April 2017 it was agreed that the RPV intended for Baltic 1 and made by Atommash would be sent to Belarus for unit 2.

Commissioning work on unit 1 by Atomstroyexport commenced in April 2019. In April 2020 hot functional tests were concluded, in May the first fuel assemblies were delivered to the site by Rosatom, and in November the unit was connected to the grid. An operating licence was issued for the unit in June 2021.

Hot functional tests at unit 2 were completed in October 2021 and fuel loading started in December 2021. The unit was connected to the grid in May 2023 and was put into commercial operation on 1 November 2023. The plant operator is Belarusian NPP.

Earlier in March 2020 the IAEA concluded an integrated nuclear infrastructure review (INIR) in line with the final phase (phase 3) of its 'milestones' approach.

Belarus used INPRO's* Nuclear Energy Systems Assessment methodology covering economics, infrastructure, waste management, proliferation resistance, physical protection, environment, and safety to confirm its investment decision. The results showed that nuclear would be competitive, with overnight costs $1960/kW and levelized electricity price 5.81 cent/kWh (compared with coal $1175/kW and 6.52 cent/kWh, and gas $805/kW and 6.76 cent/kWh). The basic overnight cost of the two units was put at $6.135 billion.

* The International Project on Innovative Nuclear Reactors and Fuel Cycles (INPRO) was developed by the International Atomic Energy Agency (IAEA).

Further nuclear power capacity

In August 2012 the government said that there were no specific plans for a second nuclear power plant but the question remained on the agenda.

In May 2018 the deputy prime minister said that two more reactors would be built "in the short term". In January 2022 Rosatom said that it was working with Belarus on the possible construction of a second plant. This was confirmed by Belarus's deputy energy minister in May.

In December 2023 the country's energy minister said that a decision on further nuclear capacity would depend on an evaluation of future electricity demand.

A VVER-1000 unit was earlier being built near Minsk but construction was abandoned in 1988 after the Chernobyl accident.

In August 2024 Belarus President Alexander Lukashenko appointed a new energy minister and said a key task would be deciding whether to build a second nuclear power plant. In November 2024 the Deputy Energy Minister announced a feasibility study on building a second plant or a third unit at the existing site would be prepared in 2025.

In March 2025 President Lukashenko stated Belarus had asked Russia to build a second nuclear power plant, likely near the eastern border. In June 2025 the Energy Minister said the country was evaluating potential locations in the Grodno, Mogilev or Gomel regions for a planned radioactive waste disposal facility required by 2030.

In November 2025 Belarus decided to build a third unit at its existing Ostrovets nuclear power plant and also begin surveys for a potential second plant site in Mogilev.

International relations

Lithuania remains opposed to operation of the plant and earlier filed a complaint with the Implementation Committee of the Espoo convention (Convention on Environmental Impact Assessment in a Transboundary Context), which ruled that Belarus had violated the convention, though Belarus denies this. Since then discussions have continued, with Germany joining the Lithuanian objections. In 2014, a meeting of the parties to the convention "encouraged" Belarus to take measures to build confidence with neighbouring countries concerning the project. They also suggested that Belarus invites the IAEA to carry out a site and external events design (SEED) mission at Ostrovets. Belarus did so, and in January 2017 the IAEA team reported: "Appropriate steps have been taken to establish the design parameters of the nuclear power plant to protect it against

the worst credible external event." It said that the plant's design parameters adequately accounted for external hazards, such as earthquakes, floods and extreme weather, and for human error. It said that hazard monitoring programs were adequate and properly documented.

The following month Lithuania's energy minister threatened to boycott electricity imports from Belarus, suggesting that this would undermine the economic foundation of the project. He proposed a bill which would prohibit access to Lithuania's grid and power market by foreign countries which "operate or build" unsafe reactors. Lithuania has had no plans to import electricity from (as distinct from through) Belarus, and its 2012 energy policy involves rebuilding the Baltic states grid to be independent of the Russian/Belarus system, and to work in the European Network of Transmission System Operators (ENTSO) synchronous system.

Russia's policy for building nuclear power plants in non-nuclear weapons states is to deliver on a turnkey basis, including supply of all fuel and repatriation of used fuel for the life of the plant. The fuel is to be reprocessed in Russia and the separated wastes returned to the client country eventually.

Earlier in May 2009 the government approved a nuclear cooperation agreement with China, which included nuclear power, joint development of innovative reactor technologies, nuclear safety, radiation protection, and environmental protection, as well as radiation technologies and their applications, nuclear medicine, and radiation therapy. It created a legal basis for Chinese participation in nuclear power plant and related construction in Belarus. It followed both a 2008 proposal from China Guangdong Nuclear Power Group (CGNPC) and a 2005 cooperation agreement. Early in 2010 official announcements said that the President had proposed to cooperate with China in nuclear power, including the construction of a power plant, although Chinese official sources did not confirm this. In July 2012 the North China Power Engineering Co Ltd (NCPE) won a contract from Grodnoenergo to build a 330 kV transmission line to connect the Ostrovets power plant to the grid. This was financed 95% by a loan from China Exim Bank.

Waste management

A radioactive waste management strategy based on IAEA principles was adopted in June 2015. It builds on regulations for nuclear and radiation safety approved by the Ministry of Emergency

Situations in September 2010. High-level waste (HLW) will be stored near the plant for its lifetime (though used fuel is returned to Russia for reprocessing). Low- and intermediate-level waste (LILW) will be stored there for up to ten years before being removed to a repository. The first stage of a LILW repository is to be constructed by 2028. The strategy also considers construction of a deep geological repository for the disposal of HLW following decommissioning of the plant.

The strategy predicts that there will be about 8400 m^3 of solid LILW radioactive waste of various types and 60 m^3 of HLW arising during the 60-year operating lifetime of the twin-unit Ostrovets plant. The projected amount of solid LILW resulting from decommissioning the plant is 2050 m^3 per unit plus 85 m^3 of HLW.

In February 2023 the government announced plans to create an organization to implement a system of long-term radioactive waste storage and disposal. The organization would oversee the construction of a radioactive storage facility aimed to be built by 2030.

Public opinion

Public opinion monitoring conducted by the Ministry of Energy jointly with the National Academy of Sciences showed a sustainable growth of support to nuclear power over the period September 2005 to May 2012. The number of nuclear power supporters increased from 28.3% to 53.5% over that period, while the number of opponents decreased from 46.7% to 21%. Slight fluctuations were observed in 2012 after the accident at Fukushima-Daiichi, which triggered “a new outburst of anxiety.” An important result of the studies is the fact that they “unambiguously recorded” positive shifts in public attitude to “calm and reasoned perception”, despite the fact that Belarus was very affected by fallout from the Chernobyl accident, with resultant evacuations.

Regulation, safety and non-proliferation

A 2007 decree (see above) ensured that nuclear and radiation safety was in line with the recommendations of the International Atomic Energy Agency (IAEA). The Nuclear & Radiation Safety Department was set up as part of the Emergencies Ministry to act as the state nuclear regulator and licensing authority. A further

decree in 2017 established the Centre of Nuclear and Radiation Safety as part of the Emergencies Ministry.

Non-proliferation

Belarus joined the Non-Proliferation Treaty (NPT) in 1995, and in 2005 signed the Additional Protocol to its safeguards agreement with the IAEA.

Notes & references

a. Belarus economic data from The Observatory of Economic Complexity, Datawheel (spun out of MIT Media Lab) [Back]